

REGULAR ENGEL DYNAMICS ON UNITRIANGULAR GROUPS: EXACT FILTRATION FIBRES AND DEPTH LAYERS

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ABSTRACT. Let q be a prime power, let N be the regular upper shift, and fix $J = I + N$ in the unitriangular group $U_n(\mathbb{F}_q)$. We determine the finite dynamics of the single map $E(X) = X^{-1}J^{-1}XJ$. If γ_k denotes the matrices whose first $k - 1$ superdiagonals vanish, then $E(\gamma_k) = \gamma_{k+1}$, and every target has exactly q^{n-k} predecessors in γ_k . Consequently every iterated restricted fibre and every depth layer has a closed form. The full functional graph is one rooted in-tree with a loop at the identity, has sharp height $n - 1$, and has $(q - 1)q^{\binom{n}{2} - 1}$ vertices in its deepest layer; its zeta function is $(1 - z)^{-1}$. One proof identifies each fibre as a left coset of the regular centralizer. A second proof solves the equation $XJ = JXY$ successively by surjective discrete differences on superdiagonals. For $U_4(\mathbb{F}_q)$, replacing J by $I + E_{12} + E_{34}$ produces image size q^2 instead of q^3 , so regularity is essential. Lang-map terminology, Engel sinks, regular centralizers, lower-central descent, and generic zeta bookkeeping receive zero contribution credit. Novelty, priority, and external circulation remain on **HOLD**.

1. INTRODUCTION AND CLAIM BOUNDARY

Fixing the second entry of a commutator turns the left Engel sequence into an ordinary finite self-map. We study that map for one specific second entry: the regular unitriangular shift. Nilpotence already says that every orbit eventually reaches the identity. The question here is finer: which lower-central levels occur as exact images, how many predecessors does each target have at every time, and what rooted tree do these data determine?

Several ingredients are owned background. The expression $X \mapsto X^{-1}\phi(X)$ is a Lang-type twisted coboundary; Lang and Steinberg are foundational sources for that setting [5, 6]. Their global surjectivity theorems do not apply to the inner automorphism used below, so we use only the terminology and the elementary fixed-coset observation. Fixed-second-variable iterated commutators are the classical left Engel sequence, and Engel sinks are studied explicitly in [1, 4]. The lower-central filtration of a unitriangular group, the centralizer of a regular Jordan block, and broader conjugacy questions in maximal unipotent groups are also established territory [3, 6]. Each of these facts receives zero contribution credit.

After that subtraction, this note records one exact conjunction for the literal fixed element $J = I + N$:

- (1) every restricted map $E : \gamma_k \rightarrow \gamma_{k+1}$ is onto with uniform fibre q^{n-k} ;
- (2) every restricted iterated fibre, cumulative depth set, exact depth layer, and sharp terminal layer is explicit;
- (3) filtration strata give an exact multitype predecessor census, while the whole graph has one recurrent point and one weak component;
- (4) the one-step theorem has a centralizer-coset proof and a separate triangular superdiagonal proof; and
- (5) a near-regular U_4 example shows that the fixed regular hypothesis cannot be widened to arbitrary noncentral unipotent elements.

Internal firewall. The closest internal proof shape is P109, where a regular nilpotent linear map sends a subspace to its image on the full subspace lattice. Its phase points are subspaces and its fibres use Grassmannian intersection counts. Here the phase points are group elements, the update is a nonlinear fixed commutator, and the fibres are centralizer cosets. P111 uses random positive products in the integer Heisenberg group and studies a word-area cocycle; neither its update nor its observables occur here. The generic slogans “nilpotent descent,” “unique absorber,” and “uniform fibre” therefore carry no credit. A bounded search found no source stating the complete (q, n, k, t) census below, but a search miss is not evidence of novelty or priority. External use remains on **HOLD**.

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2. THE FIXED REGULAR MAP AND ITS FILTRATION

Let q be a prime power and $n \geq 2$. Write E_{ij} for the usual matrix unit and put

$$N = \sum_{i=1}^{n-1} E_{i,i+1}, \quad J = I + N, \quad U_n(q) = U_n(\mathbb{F}_q).$$

Thus $U_n(q)$ is the group of upper triangular matrices with diagonal entries one. We use the commutator convention

$$(1) \quad [X, J] = X^{-1}J^{-1}XJ$$

and define the self-map

$$E : U_n(q) \longrightarrow U_n(q), \quad E(X) = [X, J].$$

For $1 \leq k \leq n$, let $\mathbf{n}_k$ be the strictly upper triangular matrices whose entries vanish on superdiagonals $1, \dots, k-1$, and set

$$\gamma_k = I + \mathbf{n}_k.$$

Then $\gamma_1 = U_n(q)$, $\gamma_n = \{I\}$, and

$$(2) \quad |\gamma_k| = q^{\sum_{r=k}^{n-1} (n-r)} = q^{(n-k)(n-k+1)/2}.$$

Since $\mathbf{n}_r \mathbf{n}_s \subseteq \mathbf{n}_{r+s}$, the inverse series for $I + A$ gives

$$(3) \quad [\gamma_r, \gamma_s] \subseteq \gamma_{r+s} \quad \text{whenever } r + s \leq n.$$

In particular, $E(\gamma_k) \subseteq \gamma_{k+1}$.

For $X \in U_n(q)$ define its entry depth

$$(4) \quad \tau(X) = \min\{t \geq 0 : E^t(X) = I\}.$$

The minimum is finite by [theorem 5.1](#). Empty sums below are zero, E^0 is the identity map, and every target is understood as an element of the full phase $U_n(q)$ unless a smaller set is displayed.

Remark 2.1 (The $n = 1$ and $n = 2$ boundaries). For $n = 1$ the phase is the singleton $\{I\}$, so the depth is zero and the zeta function is $(1 - z)^{-1}$. We state the nontrivial filtration formulas for $n \geq 2$. When $n = 2$, $U_2(q)$ is abelian, E is constant at I , and the layers have sizes 1 and $q - 1$, exactly as the general formulas give. No $n = 0$ matrix convention is used.

3. EXACT ONE-STEP FIBRES: THE CENTRALIZER ROUTE

Theorem 3.1 (Restricted image and fibre theorem). *For every $1 \leq k < n$,*

$$(5) \quad E(\gamma_k) = \gamma_{k+1}.$$

More precisely, for $Y \in U_n(q)$,

$$(6) \quad \#\{X \in \gamma_k : E(X) = Y\} = \begin{cases} q^{n-k}, & Y \in \gamma_{k+1}, \\ 0, & Y \notin \gamma_{k+1}. \end{cases}$$

Every nonempty fibre is a left coset of $C_{\gamma_k}(J)$.

Proof route I: fixed cosets and the regular centralizer. Let $\phi(X) = J^{-1}XJ$. Normality of γ_k makes ϕ an automorphism of γ_k , and $E(X) = X^{-1}\phi(X)$. If $E(X_1) = E(X_2) = Y$, then $\phi(X_i) = X_iY$ and hence

$$\phi(X_2X_1^{-1}) = X_2Y(X_1Y)^{-1} = X_2X_1^{-1}.$$

Conversely, if $h \in \gamma_k$ satisfies $\phi(h) = h$, then $E(hX) = E(X)$. Thus every nonempty fibre is exactly a left coset $C_{\gamma_k}(J)X$.

We compute this centralizer without importing an orbit formula. Commuting with J is equivalent to commuting with N . For a matrix M , the equation $MN = NM$ reads

$$M_{i,j-1} = M_{i+1,j}$$

where the boundary entries are zero. The boundary equations force M to be upper triangular, and the displayed recurrence makes each superdiagonal constant. Hence the full matrix centralizer is $\mathbb{F}_q[N]$. Intersecting its unipotent units with γ_k gives

$$(7) \quad C_{\gamma_k}(J) = \{I + a_k N^k + \dots + a_{n-1} N^{n-1} : a_i \in \mathbb{F}_q\}, \quad |C_{\gamma_k}(J)| = q^{n-k}.$$

The coset result therefore makes every nonempty fibre have size q^{n-k} . Together with the inclusion $E(\gamma_k) \subseteq \gamma_{k+1}$, [equation \(2\)](#) gives

$$|E(\gamma_k)| = \frac{|\gamma_k|}{q^{n-k}} = q^{(n-k-1)(n-k)/2} = |\gamma_{k+1}|.$$

The inclusion is equality, proving both assertions. $\square$

The left-coset orientation in [theorem 3.1](#) matters: for our convention (1), equality of two fibres fixes $X_2X_1^{-1}$ rather than $X_1^{-1}X_2$. The verifier checks the literal left coset for every target in its exhaustive lanes.

4. A SECOND ROUTE THROUGH SUPERDIAGONAL EQUATIONS

The next proof reaches [theorem 3.1](#) without the centralizer theorem, orbit-stabilizer, or a cardinality comparison.

Proposition 4.1 (Triangular solution of every fibre). *Fix $1 \leq k < n$ and $Y \in \gamma_{k+1}$. The equation $E(X) = Y$ has exactly q^{n-k} solutions $X \in \gamma_k$.*

Proof route II: successive discrete differences. Write $X = I + A$ and $Y = I + B$. From (1), the equation $E(X) = Y$ is equivalent to $XJ = JXY$. Expanding both sides and cancelling their common terms gives the exact matrix equation

$$(*) \quad AN - NA = B + NB + AB + NAB.$$

For $r \geq k$, collect the r th superdiagonal of A as

$$A^{(r)} = (a_{1,1+r}, a_{2,2+r}, \dots, a_{n-r,n}) \in \mathbb{F}_q^{n-r}.$$

On superdiagonal $r+1$, the left side of $(*)$ is

$$(8) \quad \Delta_r A^{(r)}, \quad \Delta_r(x_1, \dots, x_{n-r}) = (x_1 - x_2, \dots, x_{n-r-1} - x_{n-r}).$$

The map $\Delta_r : \mathbb{F}_q^{n-r} \rightarrow \mathbb{F}_q^{n-r-1}$ is onto: after choosing x_1 , the target differences determine $x_2, \dots, x_{n-r}$. Its kernel is the one-dimensional space of constant vectors, so every fibre has exactly q elements. For $r = n-1$, the codomain is the zero space and the same statement remains valid.

It remains to verify that these difference equations are triangular in r . The target B begins on superdiagonal $k+1$. In the equation on superdiagonal $r+1$, the term NB is fixed, while a term from AB uses a source superdiagonal u with

$$u + (k+1) \leq r+1, \quad \text{hence } u \leq r-k < r.$$

A term from NAB has $u + (k+1) + 1 \leq r+1$, and again $u < r$. Consequently the right side at stage r depends only on the prescribed B and on $A^{(k)}, \dots, A^{(r-1)}$, which have already been chosen. Starting at $r = k$, solve successively through $r = n-1$. Each of the $n-k$ stages contributes one free field coordinate, and no later equation changes an earlier choice. Every target therefore has exactly q^{n-k} solutions. $\square$

Remark 4.2 (Independence of the two routes). Route I treats E as a twisted group coboundary and proves the result from fixed cosets and a regular centralizer. Route II solves a nonlinear matrix equation by a triangular family of additive difference maps. The second route does not use the centralizer, cosets, or domain/image cardinalities. The later temporal corollaries share the one-step theorem and are not advertised as having two independent derivations.

5. ALL ITERATED FIBRES AND EXACT DEPTH LAYERS

For $0 \leq t \leq n-k$, put

$$(9) \quad S_{k,t} = \sum_{j=k}^{k+t-1} (n-j), \quad S_{k,0} = 0.$$

Theorem 5.1 (Complete restricted temporal census). *Let $1 \leq k < n$.*

TABLE 1. Full-phase exact depth layers $L_{1,t}$. Dashes are depths beyond the sharp maximum. The machine-readable artifact also lists all restricted levels k for these lanes.

q	n	$t = 0$	$t = 1$	$t = 2$	$t = 3$	$t = 4$	$t = 5$
2	6	1	31	480	3584	12288	16384
3	5	1	80	2106	17496	39366	—
4	4	1	63	960	3072	—	—

(i) For $0 \leq t \leq n - k$,

$$(10) \quad E^t(\gamma_k) = \gamma_{k+t},$$

and for every $Y \in U_n(q)$,

$$(11) \quad \#\{X \in \gamma_k : E^t(X) = Y\} = \begin{cases} q^{S_{k,t}}, & Y \in \gamma_{k+t}, \\ 0, & Y \notin \gamma_{k+t}. \end{cases}$$

For $t \geq n - k$, the image is $\{I\}$ and the fibre over I is all of γ_k .

(ii) The cumulative and exact depth counts inside γ_k are

$$(12) \quad \#\{X \in \gamma_k : \tau(X) \leq t\} = q^{S_{k,t}}, \quad 0 \leq t \leq n - k,$$

$$(13) \quad L_{k,0} = 1,$$

$$(14) \quad L_{k,t} = (q^{n-k-t+1} - 1) q^{\sum_{j=k}^{k+t-2} (n-j)}, \quad 1 \leq t \leq n - k.$$

Here $L_{k,t} = \#\{X \in \gamma_k : \tau(X) = t\}$.

(iii) On the full phase, the sharp maximum depth is $n - 1$, and the deepest layer has

$$(15) \quad (q - 1)q^{\binom{n}{2} - 1}$$

vertices.

Proof. Part (i) follows by composing the surjections in [theorem 3.1](#). A fibre of the composition has size

$$\prod_{j=k}^{k+t-1} q^{n-j} = q^{S_{k,t}},$$

including the empty product at $t = 0$. At $t = n - k$ the target is $\gamma_n = \{I\}$ and $S_{k,n-k}$ is the exponent in [equation \(2\)](#); all later iterates remain at I .

The set in (12) is the E^t -fibre of I . Subtracting the count at time $t - 1$ gives

$$q^{S_{k,t}} - q^{S_{k,t-1}} = q^{S_{k,t-1}} (q^{n-k-t+1} - 1),$$

which proves (14). Taking $k = 1$ shows that every state has depth at most $n - 1$. The last factor in (14) is nonzero for $t = n - 1$, so the bound is attained. Substitution yields

$$L_{1,n-1} = (q - 1)q^{\sum_{j=1}^{n-2} (n-j)} = (q - 1)q^{\binom{n}{2} - 1}.$$

□

The three row sums in [table 1](#) are respectively $2^{15} = 32768$, $3^{10} = 59049$, and $4^6 = 4096$. The accompanying file `code/exact_layer_table.tsv` contains 43 rows over every valid pair (k, t) for these three (q, n) lanes; the canonical verifier rebuilds the table from [equations \(2\), \(9\) and \(14\)](#) and checks its bytes.

6. THE ROOTED COMPONENT AND FILTRATION-TYPE INDEGREES

Put $\Gamma_k^\circ = \gamma_k \setminus \gamma_{k+1}$ for $1 \leq k < n$.

Corollary 6.1 (Exact functional graph). *The functional graph of E has one weak component and one recurrent point, I . After retaining the loop $I \rightarrow I$, every other vertex lies in its rooted in-tree. For $Y \in U_n(q)$,*

$$(16) \quad \text{indeg}(Y) = \#E^{-1}(Y) = q^{n-1} \mathbf{1}_{\gamma_2}(Y).$$

The loop-suppressed root has $q^{n-1} - 1$ children; any other $Y \in \gamma_2$ has q^{n-1} children; and vertices outside γ_2 have none.

More finely, for $1 \leq k \leq n - 2$,

$$(17) \quad \#\{X \in \Gamma_k^\circ : E(X) = Y\} = q^{n-k} \mathbf{1}_{\gamma_{k+1}}(Y) - q^{n-k-1} \mathbf{1}_{\gamma_{k+2}}(Y),$$

while

$$(18) \quad \#\{X \in \Gamma_{n-1}^\circ : E(X) = Y\} = (q - 1) \mathbf{1}_{\{I\}}(Y).$$

Finally,

$$(19) \quad \#\text{Fix}(E^m) = 1 \quad (m \geq 1), \quad \zeta_E(z) = \exp\left(\sum_{m \geq 1} \frac{z^m}{m}\right) = \frac{1}{1 - z}.$$

Proof. By [theorem 5.1](#), every state reaches I and the deepest layer is nonempty. Hence there is one component, its only directed cycle is the loop at I , and its height is exactly $n - 1$. Formula (16) is [equation \(6\)](#) with $k = 1$.

For $k \leq n - 2$, subtract the number of predecessors lying in γ_{k+1} from the number lying in γ_k . Applying [equation \(6\)](#) at levels k and $k + 1$ gives (17). At the last level, all q elements of γ_{n-1} map to I , and removing the identity leaves $q - 1$. The unique recurrent point is fixed by every positive iterate, so the fixed counts and the standard Artin–Mazur conversion [2] give (19). $\square$

7. WHY THE REGULAR SHIFT IS ESSENTIAL

The preceding theorem does not extend to an arbitrary unipotent choice of the second commutator entry.

Proposition 7.1 (Near-regular failure in U_4). *Let*

$$J' = I + E_{12} + E_{34} \in U_4(q)$$

and set $E'(X) = X^{-1}(J')^{-1}XJ'$. Then

$$(20) \quad |C_{U_4(q)}(J')| = q^4, \text{quad}|E'(U_4(q))| = q^2 < q^3 = |\gamma_2|.$$

Thus $E'(U_4(q)) \neq \gamma_2$, and every nonempty full-phase fibre has q^4 , not q^3 , elements.

Proof. Write

$$A = a_{12}E_{12} + a_{13}E_{13} + a_{14}E_{14} + a_{23}E_{23} + a_{24}E_{24} + a_{34}E_{34}.$$

Since $X = I + A$ commutes with J' precisely when A commutes with $E_{12} + E_{34}$, direct multiplication gives the two independent conditions

$$a_{23} = 0, \quad a_{24} = a_{13}.$$

Four coordinates remain free, proving the centralizer count. The same fixed-coset argument used in [theorem 3.1](#) applies to E' and gives $|E'(U_4(q))| = q^6/q^4 = q^2$. Every commutator in $U_4(q)$ lies in γ_2 , whose order is q^3 , so the inclusion is strict. The fibre claim follows from the same coset calculation. $\square$

This example misses only the middle simple-root entry of the regular shift. It is therefore a structural guard, not the vacuous central choice $J' = I$. The title, abstract, and all theorem statements deliberately keep the fixed regular $J = I + N$ hypothesis.

8. EXACT CONTROLS AND CONCLUSION

The standard-library verifier constructs literal polynomial-basis models of $\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_4, \mathbb{F}_5, \mathbb{F}_8$, and $\mathbb{F}_9$. It exhausts 55,808 regular phase states, checks 39 restricted surjections and 112 iterated-fibre profiles, and separately exhausts 20,514 states in the U_4 counterexample lanes. Its 1,491,877 executed assertions include statewise matrix inverses, the equation $XJ = JXY$, literal left centralizer cosets, all depth histograms, all filtration-stratum indegrees, unique periodicity through several iterates, and the exact table artifact. These finite computations are falsification controls, not proofs or owner certificates.

For the fixed regular shift, the whole temporal problem reduces exactly to one uniform surjection at each lower-central level. The centralizer route explains the fibre as a group coset; the triangular route explains each of its $n - k$ free coordinates. Multiplying those fibres yields every transient layer and the complete rooted functional graph. No statement is made for a general unipotent second entry, for another Engel word, or for conjugacy dynamics. Lang theory, Engel theory, regular centralizers, lower-central filtrations, and generic finite-map zeta identities retain zero contribution credit. Novelty, priority, specialist clearance, and external circulation remain on **HOLD**.

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